

NEURAL NETWORK PREDICTIONS OF IRANIAN STOCK INDEX USING MACROECONOMIC VALUES

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Prediction in financial affairs especially in securities is highly impotent which its aim is to decline the risk. Investors, however, make wide evaluations for investing on stocks. This study addresses the prediction of cash return index in Tehran Stock Exchange (TSE) by using Arbitrage multiple regression model and artificial neural networks. The variables included in this research are 971 samples of four daily macro economic variables divided between train and test data, namely TSE dividend and Price Index (TEDPIX), gold price, currency exchange rate (Rials/US\$) and the amount of daily transactions from 2002 to 2006.

To process Arbitrage pricing model, multi-factor regression and to process artificial neural networks (ANNs), Perceptron architecture model with two hidden layers and back-propagation algorithm with sigmoid conversion functions are applied.

To assess the performance of both models, mean absolute deviation (MAD), mean square error (MSE), mean absolute percentage error (MAPE) and root mean square error (RMSE) are utilized.

The findings show the success of both models in predicting cash return index as well as the superiority of artificial neural network over Arbitrage multiple model.

Keywords

stock return prediction, artificial neural networks, Arbitrage pricing theory